

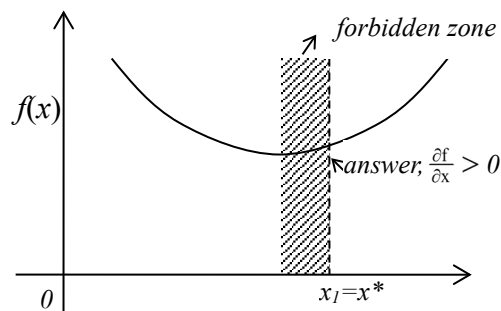
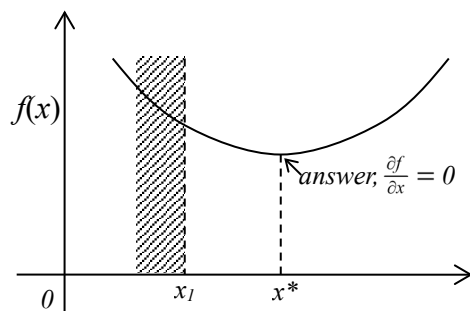
Pontryagin's Minimum Principle and Constraints

So far, we have assumed that the admissible controls and states are not constrained by any boundaries; however, in realistic systems such constraints do commonly occur. Physically realizable controls generally have magnitude limitations. State constraints often arise because of safety or structural restrictions.

If $\delta \vec{u}(t)$ is arbitrary $\rightarrow \delta J(\vec{u}^*, \delta \vec{u}) = 0$, $\vec{u}^*$ is optimal control.

If $\delta \vec{u}(t)$ is not arbitrary $\rightarrow \delta J(\vec{u}^*, \delta \vec{u}) = ?$

ex.) Min $f(x)$ function
subject to $x \geq x_1$



The first order necessary condition for minimizing $f(x)$;

$$\frac{\partial f}{\partial x} \geq 0$$

Control constraints

$$\dot{\vec{x}} = \vec{f}(\vec{x}(t), \vec{u}(t), t)$$

$$J = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t) dt$$

Subject to $\vec{x}(t_0) = \vec{x}_0, \vec{x}(t_f) = \text{free}$

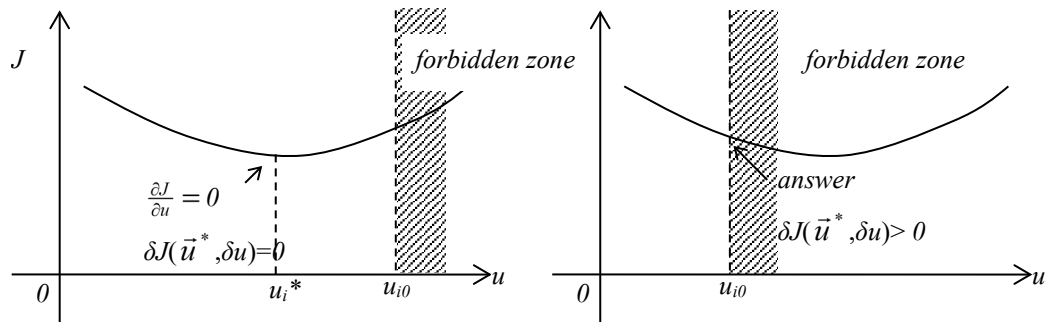
$$|u_i| \leq u_{i0}$$

Assume that boundary conditions are satisfied.

$$\delta J = \left[\frac{\partial h}{\partial \bar{x}} + \bar{v}^T \frac{\partial \bar{m}}{\partial \bar{x}} - \bar{\Lambda}^T \right]_{t=t_f} \delta \bar{x}_f + \left[\frac{\partial h}{\partial t} + \bar{v}^T \frac{\partial \bar{m}}{\partial t} + H \right]_{t=t_f} \delta t_f \\ + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \bar{u}} \delta \bar{u} + \left(\frac{\partial H}{\partial \bar{x}} + \dot{\bar{\Lambda}}^T \right) \delta \bar{x} + \delta \bar{\Lambda}^T \left(\frac{\partial H}{\partial \bar{\Lambda}} - \dot{\bar{x}} \right) \right] dt$$

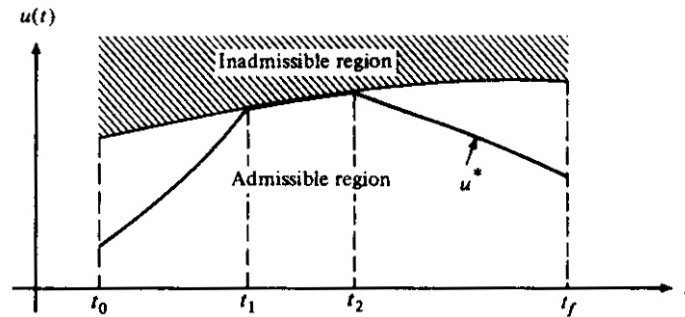
Let the state and costate differential equations be satisfied.

$$\delta J = \int_{t_0}^{t_f} \frac{\partial H}{\partial \bar{u}} \delta \bar{u} dt$$

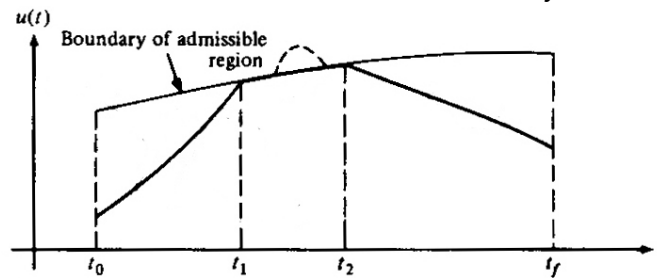


Then, $\delta J \geq 0$ for minimum J

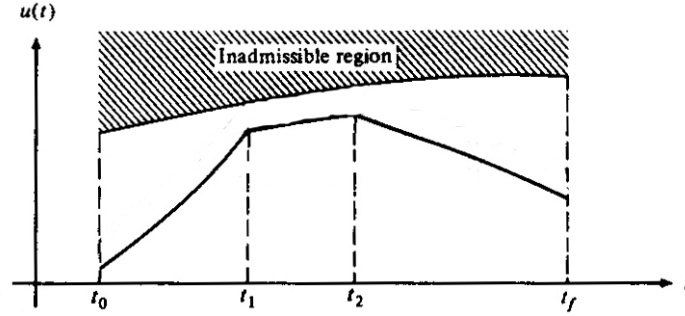
Case a) $\delta J = 0$: extremal control that lies on a boundary



Case b) $\delta J > 0$: extremal control that is constrained by a boundary



Case c) $\delta J = 0$: extremal control lies within the boundary during any portion of time interval $[t_0, t_f]$



$\Rightarrow$ Thus, case a), b), c) yields $\delta J(u^*, \delta u) \geq 0$ for the problem with control constraints.

$$\begin{aligned} \Delta H &= H(\bar{x}^*, \bar{u}^* + \delta \bar{u}, \bar{\Lambda}^*, t) - H(\bar{x}^*, \bar{u}^*, \bar{\Lambda}^*, t) \\ &= H(\bar{u}^*) + \frac{\partial H}{\partial \bar{u}} \Big|_{\bar{u}^*} \delta \bar{u} + \frac{1}{2} \frac{\partial^2 H}{\partial \bar{u}^2} \Big|_{\bar{u}^*} (\delta \bar{u})^2 + (H.O.T.) - H(\bar{u}^*) \\ &\cong \frac{\partial H}{\partial \bar{u}} \Big|_{\bar{u}^*} \delta \bar{u} \quad \delta u \text{ is sufficiently small} \\ &\quad \rightarrow (\delta u)^2 \text{ can be neglect!!} \end{aligned}$$

$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} \frac{\partial H}{\partial \bar{u}} \delta \bar{u} dt \\ &\approx \int_{t_0}^{t_f} \Delta H dt \\ &= \int_{t_0}^{t_f} \left[H(\bar{x}^*, \bar{u}^* + \delta \bar{u}, \bar{\Lambda}^*, t) - H(\bar{x}^*, \bar{\Lambda}^*, \bar{u}^*, t) \right] dt \\ &\geq 0 \end{aligned}$$

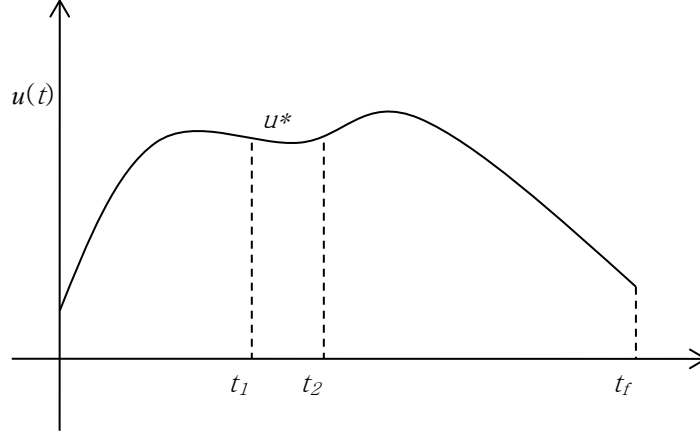
For all admissible $\delta \bar{u}$, this can be happen, if

$$\therefore H(\bar{x}^*, \bar{u}^* + \delta \bar{u}, \bar{\Lambda}^*, t) \geq H(\bar{x}^*, \bar{u}^*, \bar{\Lambda}^*, t) \text{ at every time.}$$

Proof.) Contradict the above result.

$$H(x^*, u^* + \delta u, \Lambda^*, t) < H(x^*, u^*, \Lambda^*, t) \text{ in } [t_1, t_2]$$

where $[t_1, t_2]$ is an arbitrary small, but nonzero, time interval.



$$u = u^* \quad 0 \leq t < t_1$$

$$u = u^* \quad t_2 < t < t_f$$

$$u = u^* + \delta u \quad t_1 \leq t \leq t_2$$

we can make

$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} [H(x^*, \Lambda^*, u, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &= \int_{t_0}^{t_1} [H(x^*, \Lambda^*, u^*, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &\quad + \int_{t_1}^{t_2} [H(x^*, \Lambda^*, u, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &\quad + \int_{t_2}^{t_f} [H(x^*, \Lambda^*, u^*, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &= \int_{t_1}^{t_2} [H(x^*, \Lambda^*, u, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &< 0 \end{aligned}$$

This violate the optimality condition; $\delta J \geq 0$, since $[t_1, t_2]$ can be anywhere in $[0, t_f)$ it is clear that if $H(x^*, \Lambda^*, u^* + \delta u, t) < H(x^*, \Lambda^*, u^*, t)$ for any $t \in [0, t_f)$ then, it is always possible to construct an admissible control which makes $\Delta J < 0 (\approx \delta J < 0)$, thus contracting the optimality of the control $u^*(t)$ ($\delta J \geq 0$).

Thus a necessary condition for $\vec{u}^*$ to minimize the functional J is

$$\therefore H(\vec{x}^*, u^* + \delta u, \vec{\Lambda}^*, t) \geq H(\vec{x}^*, u^*, \vec{\Lambda}^*, t)$$

for all $t \in [0, t_f]$ and for all admissible control.

This means that an optimal control must minimize the Hamiltonian

$\Rightarrow$ Pontryagin's minimum principle

In terms of the Hamiltonian

$$H \equiv g(\vec{x}, \vec{u}, t) + \vec{\Lambda}^T \vec{f}(\vec{x}, \vec{u}, t)$$

Necessary conditions for $\vec{u}^*(t)$ to be an optimal control are

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{\Lambda}}$$

$$\dot{\vec{\Lambda}} = -\frac{\partial H}{\partial \vec{x}}$$

$$H(\vec{x}^*(t), \vec{\Lambda}^*(t), \vec{u}(t), t) \geq H(\vec{x}^*(t), \vec{\Lambda}^*(t), \vec{u}^*(t), t)$$

$$\left[\frac{\partial h}{\partial \vec{x}} + \vec{v}^T \frac{\partial m}{\partial \vec{x}} - \vec{\Lambda}^T \right]_{t=t_f} \delta \vec{x}_f + \left[H + \frac{\partial h}{\partial t} + \vec{v}^T \frac{\partial m}{\partial t} \right]_{t=t_f} \delta t_f = 0$$

(If $\vec{u}$ is constrained, $\frac{\partial H}{\partial u} \neq 0$)

Pontryagin's minimum principle

If $\vec{x}^*(t), \vec{\Lambda}^*(t)$ and $\vec{u}^*(t)$ are extremal states, costates and controls, respectively.

Then, the optimal control (constrained or unconstrained control) must minimize the Hamiltonian.

$$\vec{u}^*(t) = \min_{\vec{u} \in \Omega} \arg H(\vec{x}^*(t), \vec{\Lambda}^*(t), \vec{u}(t), t)$$

The minimum principle can be also applied to problems in which the admissible controls are not bounded ($\Rightarrow \frac{\partial H}{\partial u} = 0$).

(Example) minimum-time problem

$$J = \int_0^{t_f} 1 dt = t_f, \quad t_f \text{ is free}$$

$$\begin{aligned} \dot{x}_1 &= x_2 & x_1(0) &= \text{given}, & x_1(t_f) &= 0 \\ \dot{x}_2 &= u & x_2(0) &= \text{given}, & x_2(t_f) &= 0 \\ |u| &\leq u_0 \end{aligned}$$

The problem is to transfer the system from initial state $\bar{x}_0$ to the origin in minimum time.

The Hamiltonian

$$H = 1 + \Lambda_1 x_2 + \Lambda_2 u$$

$$\Lambda^T f = (\Lambda_1 \quad \Lambda_2) \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix}$$

$$\left[\frac{\partial H}{\partial \bar{x}} + \nu^T \frac{\partial m}{\partial \bar{x}} - \bar{\Lambda}^T \right]_{t=t_f} \delta x_f + \left[H + \frac{\partial H}{\partial t} + \nu^T \frac{\partial m}{\partial t} \right]_{t=t_f} \delta t_f = 0$$

$$\rightarrow H(t_f) = 0$$

$$\dot{H} = 0 \quad \Rightarrow H(t) = \text{const.} \quad \Rightarrow H(t) = 0$$

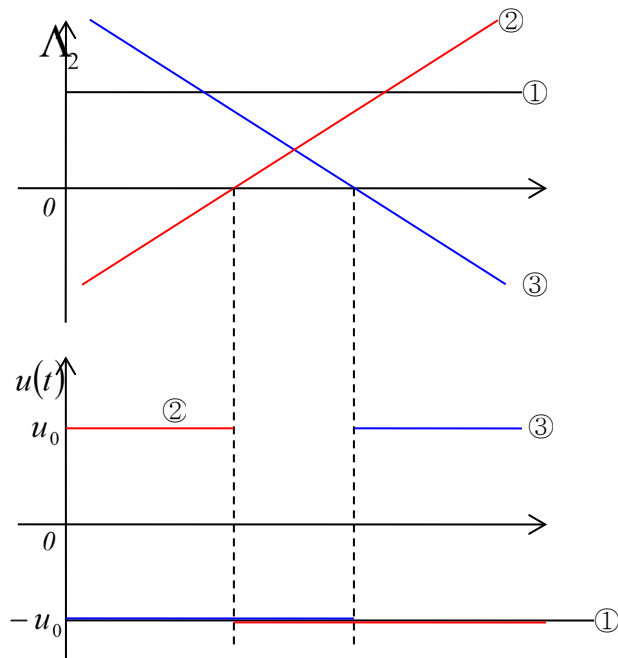
$$H(t_f) = 0$$

$$\dot{\Lambda}_1 = -\frac{\partial H}{\partial x_1} = 0 \quad \rightarrow \Lambda_1 = \text{const.}$$

$$\dot{\Lambda}_2 = -\frac{\partial H}{\partial x_2} = -\Lambda_1 \quad \rightarrow \Lambda_2 = -\Lambda_1 t + \Lambda_{20}$$

$$\bar{u}^*(t) = \min_{u \in R} \arg H(x^*(t), \Lambda^*(t), u(t), t)$$

$$\text{If } \begin{aligned} \Lambda_2 &> 0 & u &= -u_0 \\ \Lambda_2 &< 0 & u &= u_0 \end{aligned} \quad u = -u_0 \text{sign}(\Lambda_2)$$



If $u_0 = 1$

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= 1\end{aligned}$$

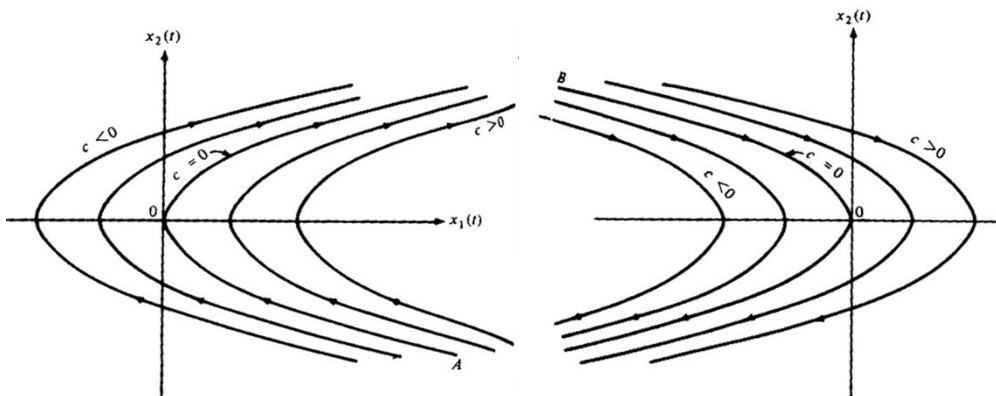
or

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -1\end{aligned} \Rightarrow \begin{aligned}dx_1 &= x_2 dt \\ dx_2 &= \pm dt\end{aligned}$$

$$\rightarrow \frac{dx_1}{dx_2} = \frac{x_2}{\pm 1}$$

$$\rightarrow \pm \int dx_1 = \int x_2 dx_2$$

$$\rightarrow \pm x_1 = \frac{1}{2} x_2^2 + c$$



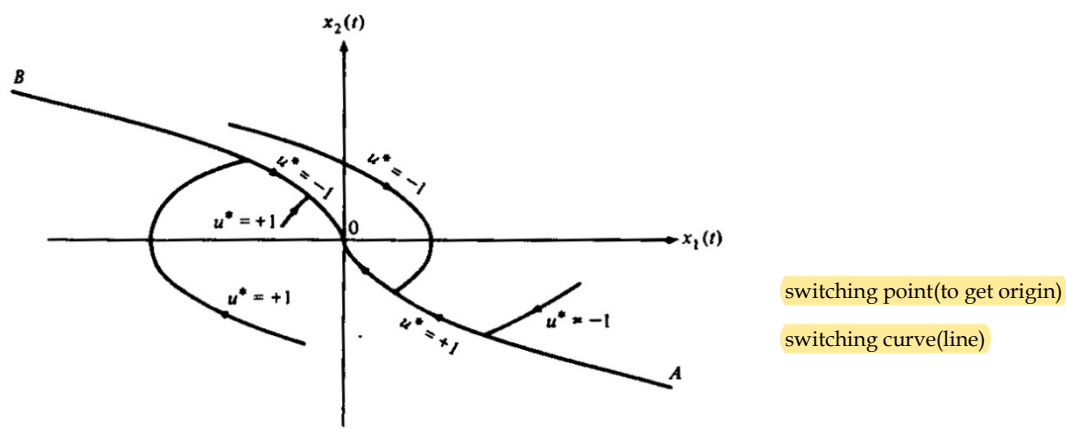
t_f is integrated time from start and to ②

$$x_1(t) = \frac{1}{2}x_2^2(t) + c_1 \quad \text{for } u = 1$$

$$x_1(t) = -\frac{1}{2}x_2^2(t) + c_2 \quad \text{for } u = -1$$

If $x_2(t) > 0$, switching curve $x_1(t) = -\frac{1}{2}x_2^2(t)$

If $x_2(t) < 0$, switching curve $x_1(t) = \frac{1}{2}x_2^2(t)$



Let $c_1 = c_2 = 0$, we find the equation of the switching curve(line)

$$x_1(t) = -\frac{1}{2}x_2(t)|x_2(t)|$$

$$s \equiv x_1 + \frac{1}{2}x_2|x_2|$$

We have obtained the **optimal control law**, that is, the optimal control at every time t is known as a function of the state variables (x_1, x_2)

Notice that

$S(\mathbf{x}(t)) > 0$ implies $\mathbf{x}(t)$ lies **above** the switching curve (A-O-B)

$S(\mathbf{x}(t)) < 0$ implies $\mathbf{x}(t)$ lies **below** the switching curve (A-O-B)

$S(\mathbf{x}(t)) = 0$ implies $\mathbf{x}(t)$ lies **on** the switching curve (A-O-B)

Thus, in terms of this switching function, the optimal control law is

$$\begin{aligned} u^*(t) &= -1, & \text{for } S(\mathbf{x}(t)) > 0 \\ &= +1, & \text{for } S(\mathbf{x}(t)) < 0 \\ &= -1, & \text{for } S(\mathbf{x}(t)) = 0 \text{ and } x_2(t) > 0 \\ &= +1, & \text{for } S(\mathbf{x}(t)) = 0 \text{ and } x_2(t) < 0 \\ &= 0, & \text{for } \mathbf{x}(t) = 0 \end{aligned}$$

< Problems with inequality constraints for a Function >

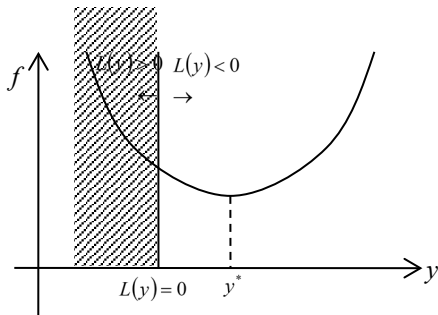
Function

Minimizing $f(y)$

Subject to $L(y) \leq 0$: inequality constraints.

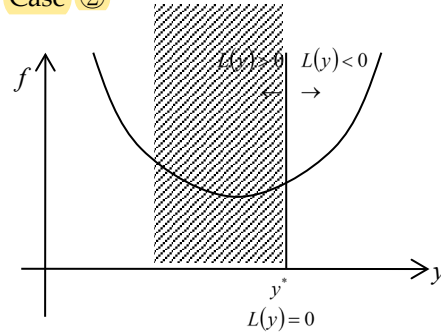
f and L are vectors of different dimension.

Case ①



$$y^* : L(y^*) < 0, \quad \frac{\partial f}{\partial y} = 0$$

Case ②



$$y^* : L(y^*) = 0, \quad \frac{\partial f}{\partial y} > 0$$

$$\rightarrow \left. \frac{\partial L}{\partial y} \right|_{y^*} < 0$$

Case ① : Constraint is not effective and can be ignored.

Case ② : Constraint is effective.

$$df = \left. \frac{\partial f}{\partial y} \right|_{y^*} dy > 0, \quad (1) \text{ } dy : \text{small perturbation about } y^*$$

$$dL = \left. \frac{\partial L}{\partial y} \right|_{y^*} dy < 0 \quad (2)$$

In order that eq(1) and eq(2) be consistent, it is clearly necessary that either.

$$\text{sign} \left(\frac{\partial f}{\partial y} \right) = -\text{sign} \left(\frac{\partial L}{\partial y} \right) \quad \text{or} \quad \frac{\partial f}{\partial y} = 0$$

These two possibilities may be expressed in one relation

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial L}{\partial y} = 0, \quad \lambda \geq 0$$

$$\text{If } \frac{\partial f}{\partial y} = 0, \quad \text{then } \lambda = 0$$

$$\frac{\partial f}{\partial y} \neq 0, \quad \text{then } \lambda > 0$$

The two cases may be treated analytically by adjoining the $f(y)$ and $L(y)$.

Define $K(y, \lambda) = f(y) + \lambda L(y)$

λ : Lagrange multiplier, inequality constraint: $L(y)$

Then, necessary conditions for minimum with respect to y becomes

$$\frac{\partial K}{\partial y} = 0 \quad \leftrightarrow \quad \frac{\partial f}{\partial y} + \lambda \frac{\partial L}{\partial y} = 0$$

and $L(y) \leq 0$

where $\lambda > 0$, $L(y) = 0$ **active** constraints, case ②

$$\left(\frac{\partial f}{\partial y} > 0, \quad \frac{\partial L}{\partial y} < 0, \quad \text{sign} \left(\frac{\partial f}{\partial y} \right) = -\text{sign} \left(\frac{\partial L}{\partial y} \right) \right)$$

$\lambda = 0$, $L(y) < 0$ **inactive** constraints, case ①

$$\left(\frac{\partial f}{\partial y} = 0, \quad \lambda = 0 \right)$$

Note: For a maximum, the sign of λ must be changed.

Note: We use this concept described in this section, to **numerically** find optimal solutions. This concept does not violate the Pontryagin's Minimum Principle.

< **Inequality constraints on the control variables** >

Inequality constraints

$$C(u, t) \leq 0 \quad (1)$$

Define $H^0 = g + \Lambda f$ (2)

$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} H_u^0 \delta u \, dt \\ &= \int_{t_0}^{t_f} \delta H^0 \, dt \end{aligned}$$

when all state and costate necessary conditions are satisfied.

We must have $\delta J \geq 0$ for all admissible δu

$$\begin{aligned} \delta H^0 &= \frac{\partial H^0}{\partial u} \delta u > 0 \\ \delta C &= \frac{\partial C}{\partial u} \delta u < 0 \quad (\text{similar to the previous section for function}) \end{aligned}$$

Define $H = g + \Lambda^T f + \mu^T C$ (3)

The necessary condition on H is

$$\frac{\partial H}{\partial u} = \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial C}{\partial u} = 0 \quad (4)$$

$$\begin{aligned} \text{where } \mu > 0 \quad C = 0 &\Rightarrow \text{eq(1) determines } u(t) \text{ at constrained arc} \\ \mu = 0 \quad C < 0 &\Rightarrow \text{eq(4) determines } u(t) \end{aligned}$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x}$$

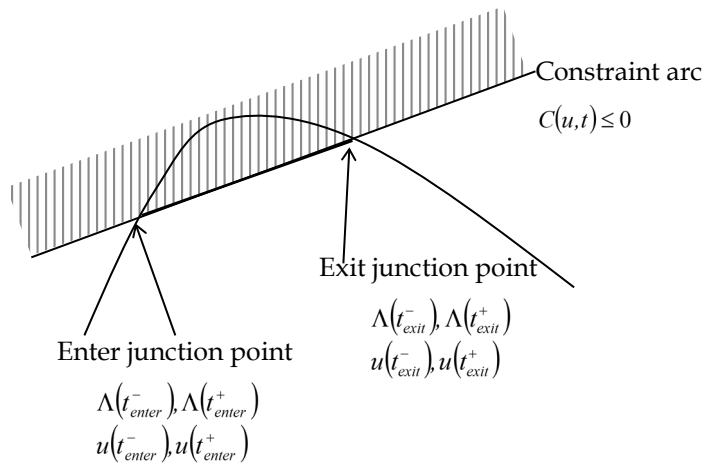
$$\dot{x} = f$$

and **Boundary Conditions**

In solving a particular problem, constrained and unconstrained arcs must be pieced together to satisfy all the necessary conditions.

At the junction points of constrained and unconstrained arcs, the control may or may not be continuous; if it is discontinuous, the point is called a corner.

There seems to be no a priori method for determining the existence of corners. If $u(t)$ is continuous at a junction point, it follows from the continuity of Λ , H_u and H that $\mu(t)$ is continuous across the junction point.



< Inequality constraints on function of the control and state variable >

Inequality constraints

$$C(x, u, t) \leq 0 \quad (1)$$

$$H = g + \Lambda^T f + \mu^T C \quad (2)$$

$$\text{where } \begin{array}{ll} \mu \geq 0 & C = 0 \\ \mu = 0 & C < 0 \end{array}$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x} - \mu^T \frac{\partial C}{\partial x}, \quad C = 0 \quad (3)$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x} \quad C < 0 \quad (4)$$

$$\dot{x} = f$$

and Boundary Conditions

$$\frac{\partial H}{\partial u} = \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial C}{\partial u} = 0 \quad (5)$$

$$\text{where } \begin{array}{ll} \mu = 0 & C < 0 \end{array} \rightarrow \text{equation (5) determines } u(t)$$

→equation (1) determines $u(t)$ at constrained arc
equation (5) determines $\mu(t)$
 μ is needed for equation (3)
eq.(3) & eq. (5) → μ, Λ for constrained arc

$$\mu \geq 0 \quad C = 0$$

< Inequality constraints on function of the state variable >

State inequality constraints

$$S(x, t) \leq 0 \quad (1)$$

We take successive total time derivative of (1) and substitute $f(x, u, t)$ for $\dot{x}$; until we obtain an expression that is explicitly dependent on u . If q^{th} time

derivatives are required, we shall call eq (1) a q^{th} -order state variable inequality constraint.

Now $S^{(q)}(x, u, t)$, the q^{th} total time derivative of S , plays the same role as $C(x, u, t)$ in previous section.

$$H = g + \Lambda^T f + \mu^T S^{(q)}(x, u, t) \quad (2)$$

where $S^{(q)} = 0$ on the constraint boundary $S = 0$
 $\mu = 0$ off the constraint boundary $S < 0$

Tangency constraint apply to the path leaving the constant boundary

$$N(x, t) \equiv \begin{Bmatrix} S(x, t) \\ S^{(1)}(x, t) \\ \vdots \\ S^{(q-1)}(x, t) \end{Bmatrix} = 0 \quad (3)$$

Equation (3) form a set of interior boundary conditions. Consequently the $\Lambda(t)$ are, in general, discontinuous at junction points between constrained and unconstrained arcs.

Necessary condtions

$$\frac{\partial H}{\partial u} = \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial S^{(q)}}{\partial u} = 0 \quad (4)$$

$$\text{where } \begin{matrix} \mu = 0 & S^{(q)} < 0 \\ \mu \geq 0 & S^{(q)} = 0 \end{matrix}$$

→ equation (4) determines $u(t)$

→ $S^{(q)} = 0$ determines $u(t)$ at constrained arc
equation (4) determines $u(t)$

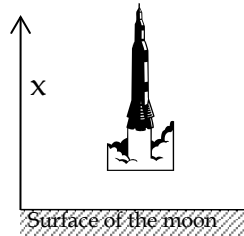
$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\left(\frac{\partial g}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} + \mu^T \frac{\partial S^{(q)}}{\partial x} \right) \quad (5)$$

$$\dot{x} = f$$

and Boundary Conditions

Example: minimum-time problem

A lunar rocket in the terminal phase of a minimum-time will make a soft landing on the surface of the moon.



m ; spacecraft mass
 $-M \leq \dot{m} \leq 0$; spacecraft mass rate, control variable
 g ; gravitational constant in surface of the moon
 T ; thrust ($T = -k\dot{m}$)
 where k is constant relative exhaust velocity of gas

$$F=ma \Rightarrow$$

$$\begin{aligned}
 m(t)\ddot{x}(t) &= -gm(t) + T(t) \\
 &= -gm(t) - k\dot{m}(t)
 \end{aligned}$$

Let $x_1 = x$, $x_2 = \dot{x}$, $x_3 = m$, $u = \dot{m}$

$$\begin{aligned}
 \dot{x}_1(t) &= x_2(t) \\
 \dot{x}_2(t) &= -g - \frac{k}{x_3(t)}u(t) \\
 \dot{x}_3(t) &= u(t)
 \end{aligned}$$

$$J = \int_{t_0}^{t_f} 1 dt$$

$$\begin{aligned}
 \rightarrow H &= 1 + \bar{\Lambda}^T \bar{f} \\
 &= 1 + \Lambda_1 x_2 + \Lambda_2 \left(-g - \frac{k}{x_3} u \right) + \Lambda_3 u \\
 &= 1 + \Lambda_1 x_2 - \Lambda_2 g + \left(-\Lambda_2 \frac{k}{x_3} + \Lambda_3 \right) u
 \end{aligned}$$

Optimal control

$$H(\Lambda^*, x^*, u^*) \leq H(\Lambda^*, x^*, u)$$

All possible $u(t)$

$$u(t) = \begin{cases} 0 & , & -\Lambda_2 \frac{k}{x_3} + \Lambda_3 < 0 \\ -M & , & -\Lambda_2 \frac{k}{x_3} + \Lambda_3 > 0 \\ \text{undetermined} & , & -\Lambda_2 \frac{k}{x_3} + \Lambda_3 = 0, \text{ for a finite time} \end{cases}$$

To obtain an explicit for $u^*(t)$, we have to solve a nonlinear two-point boundary-value problem using the numerical technique.

Minimum time problem has a “bang-bang” control effect; that is the optimal control between its maximum and minimum admissible value.

< Control variable equality constraints >

Control variable equality constraint

$$C(u, t) = 0 \quad (1)$$

$u(t)$ is the m -dimensional control vector $m \geq 2$, C is n vector function.

Hamiltonian is defined as

$$H = g + \bar{\Lambda}^T \bar{f} + \bar{\mu}^T \bar{C}$$

Optimality condition

$$\frac{\partial H}{\partial \bar{u}} = \frac{\partial g}{\partial \bar{u}} + \bar{\Lambda}^T \frac{\partial \bar{f}}{\partial \bar{u}} + \bar{\mu}^T \frac{\partial \bar{C}}{\partial \bar{u}} = 0 \quad (2)$$

Equation (1) and equation (2) represents $m + n$ equations for determining in control vector and the n function $\bar{\mu}(t)$.

All other equations remain the same.

$$\dot{\Lambda} = -\frac{\partial H}{\partial \vec{x}} = -\frac{\partial g}{\partial \vec{x}} - \Lambda^T \frac{\partial \vec{f}}{\partial \vec{x}}$$

$$\dot{x} = f$$

Boundary Conditions

< Equality constraints on functions of the control and state variables >

$$C(\vec{x}, \vec{u}, t) = 0 \quad (1)$$

$$\text{where } \frac{\partial C}{\partial \vec{u}} \neq 0 \text{ for any } u$$

$$\vec{u} \in \mathbf{R}^m$$

C can be a scalar or vector function

Hamiltonian

$$H = g + \bar{\Lambda}^T \vec{f} + \bar{\mu}^T \vec{C}$$

Optimality condition

$$\frac{\partial H}{\partial \vec{u}} = \frac{\partial g}{\partial \vec{u}} + \bar{\Lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} + \bar{\mu}^T \frac{\partial \vec{C}}{\partial \vec{u}} = 0 \quad (2)$$

Equation (1) and equation (2) represents $m + n$ equations for determining in control vector and the n function $\bar{\mu}(t)$

Costate equation

$$\dot{\Lambda}^T = -\frac{\partial H}{\partial \vec{x}} = -\frac{\partial g}{\partial \vec{x}} - \Lambda^T \frac{\partial \vec{f}}{\partial \vec{x}} - \mu^T \frac{\partial \vec{C}}{\partial \vec{x}} \quad (3)$$

All other equations remain the same.

< Equality constraints on functions of the state variables >

Equality constraint

$$S(\bar{x}, t) = 0 \quad (1)$$

$$\dot{S}(\bar{x}, \bar{u}, t) = 0$$

Equation (1) can be added a boundary condition at $t = 0$, or $t = t_f$.

Or, we can eliminate one component of $\bar{x}$ in terms of the remaining $(n-1)$ component using equation (1)

If $\dot{S}$ does not explicitly involve controls, we may take another time derivative and substitute $\dot{x} = f$, until some explicit dependence on controls does occur.

$$S^{(q)}(\bar{x}, \bar{u}, t) = 0 \quad (2) \quad (\text{qth-order state variable equality constraint})$$

plays the role of a control variable equality constraint

$$\text{where } S^{(q)} = \frac{d^q S}{dt^q}$$

$$\begin{bmatrix} S(\bar{x}, t) \\ S^1(\bar{x}, t) \\ \vdots \\ S^{(q-1)}(\bar{x}, t) \end{bmatrix} = \mathbf{0} \quad (3)$$

In addition, we must either eliminate q components of $\bar{x}$ in terms of the remaining $(n-q)$ components using equation(3).

Or add equation (3) as a set of boundary conditions $t = 0$ or $t = t_f$

$$\text{or } S(\bar{x}, t) = 0, \quad S^{(q)}(\bar{x}, \bar{u}, t) = 0$$

$$\begin{aligned} H &= g + \Lambda^T f + \mu^T S^{(q)}(x, u, t) \\ \frac{\partial H}{\partial u} &= \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial S^{(q)}}{\partial u} = 0 \\ \dot{\Lambda} &= -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x} - \mu^T \frac{\partial S^{(q)}}{\partial x} \end{aligned} \quad (4)$$

All other equations remain the same

Equation (2) and equation (4) determine $\bar{u}(t)$, $\bar{\mu}(t)$

Example 1: minimum-energy problem

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = u_1$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = u_2$$

State (equality) constraint

$$S(\vec{x}) = x_1^2 + x_3^2 - 1 = 0$$

$$J = \frac{1}{2} \int_{t_0}^{t_f} (u_1^2 + u_2^2) dt$$

t_f is fixed, $\vec{x}(t_f)$ are free, $\vec{x}(0)$ are given

Path constraint

$$\dot{S}(\vec{x}) = 2x_1\dot{x}_1 + 2x_3\dot{x}_3 = 2x_1x_2 + 2x_3x_4 = 0$$

$$\ddot{S}(\vec{x}) = 2(\dot{x}_1x_2 + x_1\dot{x}_2 + \dot{x}_3x_4 + x_3\dot{x}_4) = 2(x_2^2 + x_1u_1 + x_4^2 + x_3u_2) = 0$$

$$\rightarrow \text{constraint} \rightarrow x_2^2 + x_1u_1 + x_4^2 + x_3u_2 = 0 \quad (1)$$

Hamiltonian

$$H = \frac{1}{2}(u_1^2 + u_2^2) + \Lambda_1x_2 + \Lambda_2u_1 + \Lambda_3x_4 + \Lambda_4u_2 + \mu(x_2^2 + x_1u_1 + x_4^2 + x_3u_2)$$

Optimal control

$$\frac{\partial H}{\partial u_1} = u_1 + \Lambda_2 + \mu x_1 = 0 \quad \rightarrow u_1 = -(\Lambda_2 + \mu x_1) \quad (2)$$

$$\frac{\partial H}{\partial u_2} = u_2 + \Lambda_4 + \mu x_3 = 0 \quad \rightarrow u_2 = -(\Lambda_4 + \mu x_3) \quad (3)$$

Substitute equation (2), equation (3) $\rightarrow$ equation (1)

$$x_2^2 + x_1u_1 + x_4^2 + x_3u_2 = 0$$

$$\begin{aligned}
&\rightarrow x_2^2 + x_4^2 - x_1(\Lambda_2 + \mu x_1) - x_3(\Lambda_4 + \mu x_3) = 0 \\
&\rightarrow x_2^2 + x_4^2 - x_1\Lambda_2 - x_3\Lambda_4 - \mu(x_1^2 + x_3^2) = 0 \\
&\rightarrow \underbrace{(x_1^2 + x_3^2)}_{=1} \mu = x_2^2 + x_4^2 - x_1\Lambda_2 - x_3\Lambda_4 \\
&\rightarrow \mu = x_2^2 + x_4^2 - x_1\Lambda_2 - x_3\Lambda_4 \tag{4}
\end{aligned}$$

$$\dot{\Lambda}(t) = -\frac{\partial H}{\partial x(t)} = -\left(\frac{\partial g}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} + \mu^T \frac{\partial S^{(2)}}{\partial x} \right) \tag{5}$$

Boundary conditions

$x(0)$ given $\rightarrow$ 4 boundary conditions

$x(t_f)$ should be on $S(\vec{x}(t_f)) = 0$

$$\rightarrow x_1^2(t_f) + x_3^2(t_f) - 1 = 0$$

$$\begin{aligned}
x(t_f) \text{ is free} \rightarrow \frac{\partial h}{\partial \vec{x}}(\vec{x}(t_f), t_f) + \nu^T \frac{\partial \vec{m}}{\partial \vec{x}} - \Lambda(t_f) &= 0 \\
(h \text{ is 0 in this problem})
\end{aligned}$$

$$\Rightarrow \bar{\Lambda}(t_f) = \nu^T \frac{\partial \vec{m}}{\partial \vec{x}} = \nu^T \frac{\partial S}{\partial \vec{x}} \quad \text{at } t_f$$

The following 5 eqs. at t_f can numerically determine 4 boundary conditions($\bar{\Lambda}(t_f)$) and ν .

$$\Lambda_1(t_f) = 2\mu x_1(t_f), \quad \Lambda_2(t_f) = 0, \quad \Lambda_3(t_f) = 2\nu x_3(t_f), \quad \Lambda_4(t_f) = 0$$

$$x_1^2(t_f) + x_3^2(t_f) - 1 = 0$$

Example2: minimum-fuel problem

$$\dot{x}(t) = u(t)$$

The system is to be transferred from an arbitrary initial state x_0 to the origin.

Performance measure

$$J = \int_{t_0}^{t_f} |u(t)| dt \quad \text{where } t_f \text{ is free}$$

$$|u(t)| \leq 1.0$$

Find optimal control using the Pontryagin's Minimum principle.

Hamiltonian $H = |u(t)| + \Lambda u(t)$

Costate equation $\dot{\Lambda}(t) = -\frac{\partial H}{\partial x} = 0$
 $\Lambda(t) = C_1 (\text{const.})$

Optimal control

$$u^*(t) = \begin{cases} 1 & , & C_1 < -1 \\ 0 & , & -1 < C_1 < 1 \\ -1 & , & C_1 > 1 \\ \text{undetermined} & , & C_1 = -1 \text{ or } C_1 = 1 \text{ for a finite time} \end{cases}$$

< A weighted combination of time and fuel >

$$J = \int_{t_0}^{t_f} (w + |u(t)|) dt$$

where t_f is free

$w > 0$ is chosen to weight the relative importance of time and fuel.

Example: minimum-energy problem

$$\dot{x}(t) = -ax(t) + u(t) \quad a > 0$$

The system is to be transferred from an arbitrary initial state x_0 to the origin by a constraint minimizes the performance index.

$$J(u) = \int_{t_0}^{t_f} [w + u^2(t)] dt \quad \text{where } t_f \text{ is free, } w > 0$$

The admissible controls are constrained by

$$|u(t)| \leq 1.0$$

Find optimal control using the Pontryagin's Minimum principle.

$$\text{Hamiltonian} \quad H = w + u^2(t) - \Lambda ax(t) + \Lambda u(t)$$

$$\text{Costate equation} \quad \dot{\Lambda}(t) = -\frac{\partial H}{\partial x} = a\Lambda(t) \quad \rightarrow \quad \Lambda(t) = C_1 e^{at}$$

Unbounded control case (i.e., $|u(t)| < 1.0$)

$$\frac{\partial H}{\partial u} = 0 = 2u + \Lambda$$

$$\Rightarrow u^* = -\frac{1}{2}\Lambda \quad \leftrightarrow \quad |\Lambda(t)| < 2 \quad \text{when } |u(t)| < 1.0$$

If $|\Lambda(t)| \geq 2$, then the control that minimized H is

$$u^*(t) = \begin{cases} 1 & , \quad \Lambda(t) \leq -2 \\ -1 & , \quad \Lambda(t) \geq 2 \end{cases}$$

thus,

$$u^*(t) = \begin{cases} 1 & , \quad \Lambda(t) \leq -2 \\ -\frac{1}{2}\Lambda(t) & , \quad -2 < \Lambda(t) < 2 \\ -1 & , \quad \Lambda(t) \geq 2 \end{cases}$$

Note:

In minimum-time problems, the optimal control is “bang-bang”

In minimum-fuel problems, the optimal control is “bang-off-bang”

In minimum-energy problems, the optimal control is a continuous function.

< Singular intervals >

If there is a time interval $[t_1, t_2]$ of finite duration which the necessary condition provides no information about the relationship between $x^*(t)$, $\Lambda^*(t)$ and $u^*(t)$, then we say that the problem is singular or the control is singular control, or the state on that interval is singular arc, the interval $[t_1, t_2]$ is a singular interval.

Example 1: $J = \int_{t_0}^{t_f} 1 dt$

$$\dot{x}_1(t) = x_2(t), \quad \dot{x}_2(t) = u(t)$$

$$x_1(0), x_2(0) \text{ are given.}$$

$$x_1(t_f), x_2(t_f) \text{ are given.}$$

$$|u(t)| \leq 1.0$$

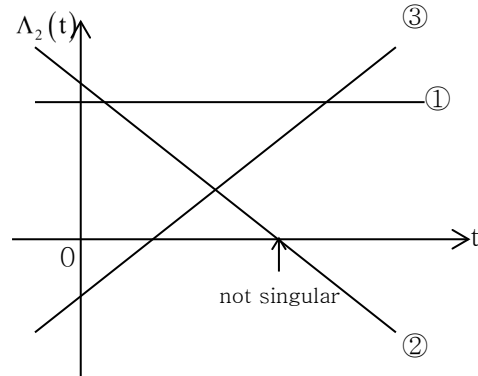
$$H = 1 + \Lambda_1 x_2 + \Lambda_2 u$$

$$u = -\text{sign}(\Lambda_2)$$

$$\dot{\Lambda}_1 = -\frac{\partial H}{\partial x_1} = 0$$

$$\dot{\Lambda}_2 = -\frac{\partial H}{\partial x_2} = -\Lambda_1$$

$$\Lambda_2(t) = -\Lambda_1 t + \Lambda_{20}$$



Singular interval cannot occur for this problem. How can we know this problem has singular control or not?

Let Λ_2 be zero for a finite time

$$\Lambda_2 = 0 \rightarrow \dot{\Lambda}_2 = 0 \Rightarrow \Lambda_1 = 0 \rightarrow H(t) = 1 : \text{this is not possible.}$$

In general,

$$\begin{aligned} H(t_f) &= -\frac{\partial h}{\partial t} - v^T \frac{\partial m}{\partial t} \quad \text{at } t = t_f \\ &= 0 \\ H(t) &= \text{const.} \\ \Rightarrow H(t) &= 0 \end{aligned}$$

So, Λ_2 cannot be zero for a finite time.

$\Rightarrow$ thus, singular intervals cannot exist.

Example 2: $J = \frac{1}{2} \int_{t_0}^{t_f} x_1^2 dt$

$$\dot{x}_1 = x_2 + u, \quad \dot{x}_2 = -u$$

$$x_1(0), x_2(0), t_f \text{ are given.} \quad x_1(t_f), x_2(t_f) \text{ are given}$$

u is unbounded

$$H = \frac{1}{2} x_1^2 + \Lambda_1(x_2 + u) - \Lambda_2 u$$

$$\dot{\Lambda}_1(t) = -\frac{\partial H}{\partial x_1} = -x_1$$

$$\dot{\Lambda}_2(t) = -\frac{\partial H}{\partial x_2} = -\Lambda_1$$

$$\frac{\partial H}{\partial u} = 0 \rightarrow \Lambda_1 - \Lambda_2 = 0$$

when we cannot determine the control from $\frac{\partial H}{\partial u} = 0$ explicitly, then the control

is said to be singular. So, singular intervals exist for this problem.

Assume, $\Lambda_1 - \Lambda_2 = 0$ for a finite time

$$\begin{aligned}\dot{\Lambda}_1 - \dot{\Lambda}_2 = 0 &\rightarrow -x_1 + \Lambda_1 = 0 &\Rightarrow \Lambda_1 = x_1 \\ \ddot{\Lambda}_1 - \ddot{\Lambda}_2 = 0 &\rightarrow -\dot{x}_1 + \dot{\Lambda}_1 = -(x_2 + u) - x_1 = 0 &\Rightarrow u = -(x_1 + x_2)\end{aligned}$$

The singular interval is called a singular arc of second order.

< Discontinuities in the state variables at interior points >

Suppose that with the general optimal programming problem, we have a set of interior boundary conditions,

$$N(\bar{x}(t_1), t_1) = 0 \quad (1)$$

Where t_1 is some intermediate time, $t_0 < t_1 < t_f$, and N is s q -component vector function. We now a three-point boundary-value problem instead of two-point boundary-value problem. Now, eq(1) represents a set of terminal constraints for the part of the path from t_0 to t_1 .

Example:

$$\dot{x} = u$$

$$x(0) = x_0 \quad x(1) = 0$$

$$J = \int_0^1 g(x, u) dt$$

Derive the necessary conditions for optimality if x is suddenly changed at a time $0 < t_1 < 1$ as follows;

$$x(t_1^+) = x(t_1^-) + 1$$

Is the Hamiltonian continuous or not?

$$J = \int_0^{t_1^-} g(x, u) dt + \int_{t_1^+}^1 g(x, u) dt$$

We can obtain necessary conditions by using Lagrange multiplier functions $\Lambda(t)$ and constant Lagrange multiplier ν .

$$J_a = \nu(x(t_1^+) - x(t_1^-) - 1) + \int_0^{t_1^-} [g(x, u) + \Lambda f - \Lambda \dot{x}] dt + \int_{t_1^+}^1 [g(x, u) + \Lambda f - \Lambda \dot{x}] dt$$

Let's define Hamiltonian $H = g(x, u) + \Lambda f$ and $\Phi \equiv \nu(x(t_1^+) - x(t_1^-) - 1)$

$$\begin{aligned} \delta J_a &= \frac{\partial \Phi}{\partial x(t_1^-)} \delta x_{t_1^-} + \frac{\partial \Phi}{\partial x(t_1^+)} \delta x_{t_1^+} + \frac{\partial \Phi}{\partial t_1} \delta t_1 \\ &+ \int_0^{t_1^-} \left[\frac{\partial H}{\partial x} \delta x + \frac{\partial H}{\partial u} \delta u + \frac{\partial H}{\partial \Lambda} \delta \Lambda - \dot{x} \delta \Lambda - \Lambda \delta \dot{x} \right] dt + (H - \Lambda \dot{x})_{t_1^-} \delta t_1 \\ &+ \int_{t_1^+}^1 \left[\frac{\partial H}{\partial x} \delta x + \frac{\partial H}{\partial u} \delta u + \frac{\partial H}{\partial \Lambda} \delta \Lambda - \dot{x} \delta \Lambda - \Lambda \delta \dot{x} \right] dt - (H - \Lambda \dot{x})_{t_1^+} \delta t_1 \end{aligned}$$

Using integration by part

$$\begin{aligned} \delta J_a &= \frac{\partial \Phi}{\partial x(t_1^-)} \delta x_{t_1^-} + \frac{\partial \Phi}{\partial x(t_1^+)} \delta x_{t_1^+} + \frac{\partial \Phi}{\partial t_1} \delta t_1 \\ &- \Lambda \delta x \Big|_0^{t_1^-} + \int_0^{t_1^-} \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt + (H - \Lambda \dot{x})_{t_1^-} \delta t_1 \\ &- \Lambda \delta x \Big|_{t_1^+}^1 + \int_{t_1^+}^1 \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt - (H - \Lambda \dot{x})_{t_1^+} \delta t_1 \end{aligned}$$

$$\text{Note : } \delta x_{t_1^\pm} = \delta x(t_1^\pm) + \dot{x}(t_1^\pm) \delta t_1 \quad \rightarrow \quad \delta x(t_1^\pm) = \delta x_{t_1^\pm} - \dot{x}(t_1^\pm) \delta t_1$$

$$-\Lambda \delta x \Big|_0^{t_1^-} \Rightarrow -[\Lambda \delta x(t_1^-) - \Lambda \delta x(0)] = -\Lambda [\delta x_{t_1^-} - \dot{x}(t_1^-) \delta t_1]$$

$$-\Lambda \delta x \Big|_{t_1^+}^1 \Rightarrow -[\Lambda \delta x(1) - \Lambda \delta x(t_1^+)] = \Lambda [\delta x_{t_1^+} - \dot{x}(t_1^+) \delta t_1]$$

$$\begin{aligned}
\delta J_a = & \left[\frac{\partial \Phi}{\partial t_l} + \Lambda \dot{x}(t_l^-) - \Lambda \dot{x}(t_l^+) + H(t_l^-) - \Lambda \dot{x}(t_l^-) - H(t_l^+) + \Lambda \dot{x}(t_l^+) \right] \delta t_l \\
& + \left[\frac{\partial \Phi}{\partial x(t_l^-)} - \Lambda(t_l^-) \right] \delta x_{t_l^-} + \left[\frac{\partial \Phi}{\partial x(t_l^+)} + \Lambda(t_l^+) \right] \delta x_{t_l^+} \\
& + \int_0^{t_l^-} \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt + \int_{t_l^+}^l \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt
\end{aligned}$$

$\delta J = 0$ by the fundamental theorem of Calculus of variation

We obtain the necessary conditions

$$\frac{\partial \Phi}{\partial t_1} + H(t_1^-) = H(t_1^+) \Rightarrow \text{if } \frac{\partial \Phi}{\partial t_1} = 0 \rightarrow H(t_1^-) = H(t_1^+)$$

$$\Lambda(t_1^-) = \frac{\partial \Phi}{\partial x(t_1^-)}$$

$$\Lambda(t_1^+) = -\frac{\partial \Phi}{\partial x(t_1^+)}$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x}$$

$$\frac{\partial H}{\partial u} = 0$$

$$\dot{x} = \frac{\partial H}{\partial \Lambda} = f$$

< Interior-point Constraints >

In some problems there will be discontinuities in the state variables as well as discontinuities of the system equations at interior points.

Furthermore, the performance index and the constraints may be functions of the state and/or time at several discrete points. A general class of such problems involves choosing $u(t)$ to minimize

$$J = \phi \left[x(t_0^-), x(t_0^+), \dots, x(t_N^-), x(t_N^+); t_0, \dots, t_N \right] + \sum_{i=1}^N \int_{t_{i-1}^+}^{t_i^-} L^{(i)}(x(t), u(t), t) dt \quad (1)$$

subject to constraints

$$\dot{x} = f^{(i)}(x(t), u(t), t) \quad , \quad t_{i-1} < t < t_i \quad i = 1, 2, \dots, N \quad (2)$$

$$\psi^{(j)} \left[x(t_0^-), x(t_0^+), \dots, x(t_N^-), x(t_N^+); t_0, \dots, t_N \right] = 0 \quad j=0, 1, 2, \dots, N \quad (3)$$

Where $x(t_i^-)$ signifies the state vector just before $t=t_i$ and $x(t_i^+)$ signifies the state vector just after $t=t_i$.

Necessary conditions for a minimum are derived by adjoining equation (2) and equation (3) to equation (1) by Lagrange multiplier function $\lambda(t)$ and constant Lagrange multiplier $\nu^{(i)}$ respectively

$$J_a = \phi + \sum_{j=0}^N \left[\nu^{(j)} \right]^T \psi^{(j)} + \sum_{i=1}^N \int_{t_{i-1}^+}^{t_i^-} \left[L^{(i)} + \lambda^T f^{(i)} - \lambda^T \dot{x} \right] dt \quad (4)$$

$$\text{Define } \Phi = \phi + \sum_{j=0}^N \left[\nu^{(j)} \right]^T \psi^{(j)} \quad (5)$$

$$H^{(i)} \equiv L^{(i)} + \lambda^T f^{(i)} \quad i = 1, 2, \dots, N \quad (6)$$

The first variation of equation (4), using the usual integration by part, is then

$$\begin{aligned}
\delta J_a = & \sum_{i=0}^N \left[\frac{\partial \Phi}{\partial t_i} \delta t_i + \frac{\partial \Phi}{\partial x(t_i^-)} \delta x_{t_i}^- + \frac{\partial \Phi}{\partial x(t_i^+)} \delta x_{t_i}^+ \right] \\
& + \sum_{i=1}^N \left\{ \left[H^{(i)} - \lambda^T \dot{x} \right]_{t=t_i^-} \delta t_i - \left[H^{(i+1)} - \lambda^T \dot{x} \right]_{t=t_i^+} \delta t_i \right\} \\
& + \sum_{i=1}^N \left\{ \left[-\lambda \delta x \right]_{t_i^-}^{t_i^+} + \int_{t_{i-1}^+}^{t_i^-} \left[\left(\dot{\lambda}^T + \frac{\partial H^{(i)}}{\partial x} \right) \delta x + \frac{\partial H^{(i)}}{\partial u} \delta u + \left(\frac{\partial H^{(i)}}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt \right\} \\
= & 0
\end{aligned} \tag{7}$$

Eliminating $\delta x(t_i^\pm)$ from equation (7) using the relations

$$\delta x_{t_i}^\pm = \delta x(t_i^\pm) + \dot{x}(t_i^\pm) \delta t_i, \quad \delta x(t_i^\pm) = \delta x_{t_i}^\pm - \dot{x}(t_i^\pm) \delta t_i \tag{8}$$

We obtain by regrouping

$$\begin{aligned}
\partial J_a = & \sum_{i=0}^N \left[\frac{\partial \Phi}{\partial t_i} + H^{(i)}(t_i^-) - H^{(i+1)}(t_i^+) + \Lambda \dot{x}(t_i^-) - \Lambda \dot{x}(t_i^+) - \Lambda \dot{x}(t_i^-) + \Lambda \dot{x}(t_i^+) \right] \delta t_i \\
& + \sum_{i=1}^N \left\{ \left[\frac{\partial \Phi}{\partial x(t_i^-)} - \lambda^T(t_i^-) \right] \delta x_{t_i}^- \right\} + \sum_{i=0}^{N-1} \left\{ \left[\frac{\partial \Phi}{\partial x(t_i^+)} + \lambda^T(t_i^+) \right] \delta x_{t_i}^+ \right\} \\
& + \sum_{i=1}^N \int_{t_{i-1}^+}^{t_i^-} \left[\left(\dot{\lambda}^T + \frac{\partial H^{(i)}}{\partial x} \right) \delta x + \frac{\partial H^{(i)}}{\partial u} \delta u + \left(\frac{\partial H^{(i)}}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt = 0
\end{aligned} \tag{9}$$

where it is to be noted that $H^{(0)} = H^{(N+1)} \equiv 0$

We obtain the following necessary conditions

$$\dot{\lambda}^T = -\frac{\partial H^{(i)}}{\partial x}, \quad , \quad t_{i-1}^+ < t < t_i^- \quad , \quad i=1, \dots, N \tag{10}$$

$$\lambda^T(t_i^-) = \frac{\partial \Phi}{\partial x(t_i^-)}, \quad , \quad i = 1, \dots, N \tag{11}$$

$$\lambda^T(t_i^+) = -\frac{\partial \Phi}{\partial x(t_i^+)}, \quad , \quad i = 0, \dots, N-1 \tag{12}$$

And let us choice $t_i (i = 0, \dots, N)$ to satisfy (transvrsality conditions)

$$\frac{\partial \Phi}{\partial t_i} + H^{(i)}(t_i^-) - H^{(i+1)}(t_i^+) = 0 \quad , \quad i = 0, \dots, N \quad (13)$$

If t_i is specified, the corresponding relation in equation (13) is not necessary since $\delta t_i = 0$ in equation (9). Similarly, if $x(t_0^+)$ is specified, we have $\delta x_0^+ = 0$ in equation (9) and equation (12), with $i = 0$ is not necessary.

We must also choose $\nu^{(j)}$ to satisfy the conditions $\psi^{(j)} = 0$ in equation (13).

Again, an optimality condition must be made to justify setting

$$\frac{\partial H^{(i)}}{\partial u} = 0 \quad (14)$$

in equation (9) since $\delta u(t)$ is arbitrary but must produce δx_i^- , δx_i^+ , δt_i consistent with

$$\delta \psi^{(j)} = \sum_{i=0}^N \left[\frac{\partial \psi^{(j)}}{\partial t_i} \delta t_i + \frac{\partial \psi^{(j)}}{\partial x(t_i^-)} \delta x_i^- + \frac{\partial \psi^{(j)}}{\partial x(t_i^+)} \delta x_i^+ \right] = 0 \quad , \quad j = 0, \dots, N \quad (15)$$

Equations (10), (11), (12) and (14) are necessary conditions and equation (13) is a set of transversality conditions.

< Hamilton's principle in Mechanics >

The motion of a conservative system, from time t_0 to t_f is such that the integral

$$J = \int_{t_0}^{t_f} L(u, q) dt \quad (1)$$

has a stationary value(=Hamiltonian principle)

where $L = T(u, q) - V(q)$ = the Lagrangian of the system

T = kinetic energy of the system

V = potential energy of the system

q = generalized coordinate vector (state of the system)

$\dot{q} = u$ = generalized velocity vector.

Optimal control theory

The Hamiltonian: $H = L + \Lambda^T f \quad (2)$

Consequently, the Euler-Lagrange equations are

$$\dot{\Lambda}^T = -\frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q} \quad (3)$$

$$0 = \frac{\partial H}{\partial u} = \frac{\partial L}{\partial u} + \Lambda^T \quad (4)$$

Nature chooses $\dot{q}(=u)$ as a control value.

combining these last two vector equations

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial u} \right) + \frac{d}{dt} (\Lambda^T) \quad (5)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \quad (6)$$

Lagrange equations of motion for a conservation system.

if L is not explicit function of time, the Hamiltonian is constant.

$$\begin{aligned}
 H &= L + A^T u \\
 &= L - \frac{\partial L}{\partial u} u \\
 &= T - V - \frac{\partial L}{\partial u} u \\
 &= \text{const.}
 \end{aligned}$$

T is a homogeneous quadratic form in u ($T = \frac{1}{2}mv^2 = \frac{1}{2}m\dot{q}^2 = \frac{1}{2}mu^2$)

$$\frac{\partial T}{\partial u} u = 2T$$

Hence, we have

$$\begin{aligned}
 H &= T - V - 2T \\
 &= -(T + V) \\
 &= \text{const.}
 \end{aligned}$$

That is, the kinetic energy plus potential energy is constant during the time.